

Queue Implemented as a Linked List

Due at the start of your next lab period

Objectives

Use pointers.

Use dynamic memory allocation.

Maintain a dynamically-allocated data structure.

Requirements

Write a C program to interactively build and maintain a dynamically linked list, specifically a queue.

Your program must:

- 1) Interact with the user to build a dynamically-allocated singly-linked queue.
- 2) Correctly handle the empty condition.
- 3) Consist of three files: `queueMain.c`, `queue.c`, and `queue.h`, where I have supplied `queue.h` and `queueMain.c` on the course web site. You just need to add the functions that are declared in `queue.h` and write the code in `queue.c`
- 4) Document `queueMain.c`

Hints

In a queue, new items added to the rear of the queue whereas items are removed from the front of the queue.

With the optional double-ended queue (or deque), additions and removals can be done from either end. To do this, you need to implement a doubly-linked list (modify `queue.h`) and modify my `main()` (to prompt the user with additional options).

Note that `queueMain.c` never uses the data type `ListNode`. This is as it should be: the user of the Queue should neither need to know nor care about how the Queue is implemented.

Evaluation

A working, well-documented program to the instructor 9/10

Optional: Implement a double-ended queue (deque) using a doubly-linked list 1/10

Sample Runs

```
dale$ ./queue
Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 0
Enter an integer: 1
Added 1
Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 0
Enter an integer: 2
Added 2
Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 0
Enter an integer: 3
Added 3
Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 0
Enter an integer: 4
Added 4
Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 0
Enter an integer: 7
```

Added 7

Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 1

Removed 1

Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 1

Removed 2

Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 2

3 Items currently in the queue:

3

4

7

Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 1

Removed 3

Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): -1

2 Items remaining in the queue:

4

7

dale\$./queue

Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 0

Enter an integer: 4

Added 4

Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 1

Removed 4

Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 1

Queue is empty

Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 0

Enter an integer: 5

Added 5

Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 2

1 Items currently in the queue:

5

Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): 0

Enter an integer: 7

Added 7

Enter 0 to add to queue, 1 to remove, or 2 to print contents (-1 to quit): -1

2 Items remaining in the queue:

5

7